

Advanced International Macroeconomics B (EI136)

Class 11

Macroeconomics of tariffs

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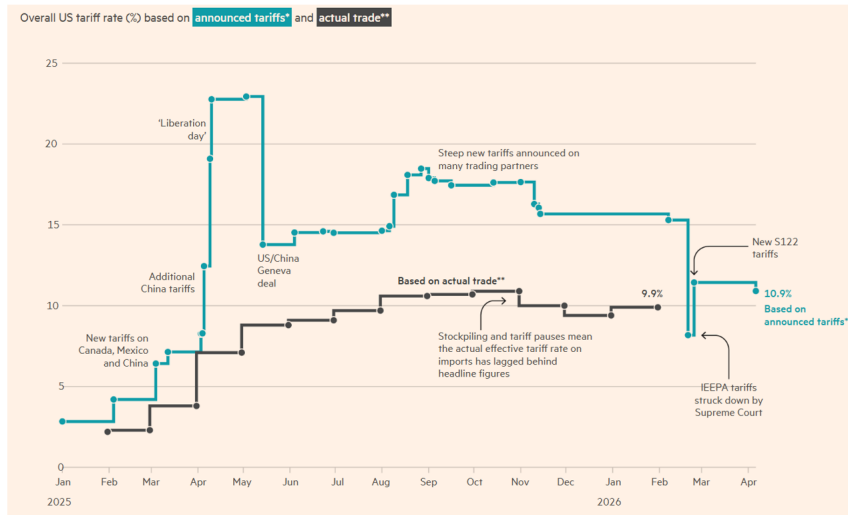
May 11, 2026

Outline of today class

- Impact of tariffs on the exchange rate (Ostry, Lloyd, and Corsetti 2025).
 - Control for retaliations by other countries.
- Macro impact of tariff in a model (Auray, Devereux, and Eyquem 2025).
 - Imported inputs and monetary policy.
- Optimal monetary policy responses (Bianchi, and Coulibaly 2025).
 - Accept some inflation to handle a fiscal revenue externality.

Relevance of tariffs

- Rapid increase in 2025, but also earlier trade war rounds in 2018-2020.



<https://www.ft.com/content/2c473393-35fb-479d-8bba-236a1a98087c?syn-25a6b1a6=1>

IMPACT ON EXCHANGE RATE

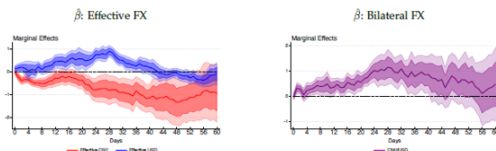
Control for retaliation

- A tariff on imports improves competitiveness, leading to a rebalancing through an exchange rate appreciation.
 - In April 2025 the dollar however depreciated.
- Controls for expected retaliation by other countries (Ostry, Lloyd, and Corsetti 2025).
- Database of tariff announcements by the U.S. (controlling for magnitude) in 2018-20 and 2025.
 - Distinguish between tariffs followed by retaliations and tariffs without retaliation.
 - 2018-20: can look at the impact of Chinese and European tariffs.
- A tariff without retaliation appreciate the dollar.
- A tariff with retaliation depreciates the dollar (retaliation more than offsets).
 - With dollar pricing, tighter monetary policy abroad than in the U.S., leads to a weaker dollar.

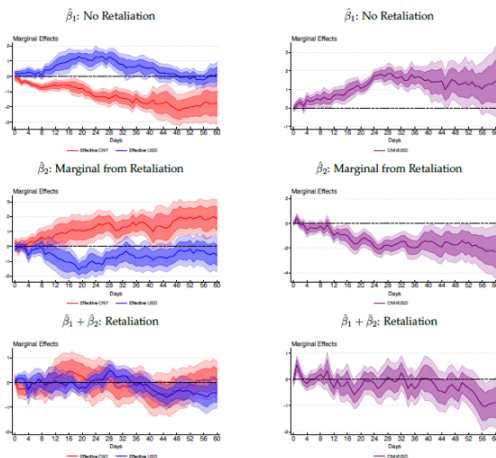
- US-China: US tariff without retaliation **appreciates** the effective dollar and **depreciates** the RMB, also **appreciates** the dollar bilaterally.
- No effect with retaliation.

Figure 3: Estimated Impacts of 2018-2020 Tariff Events on Exchange Rates

A. Average Impacts



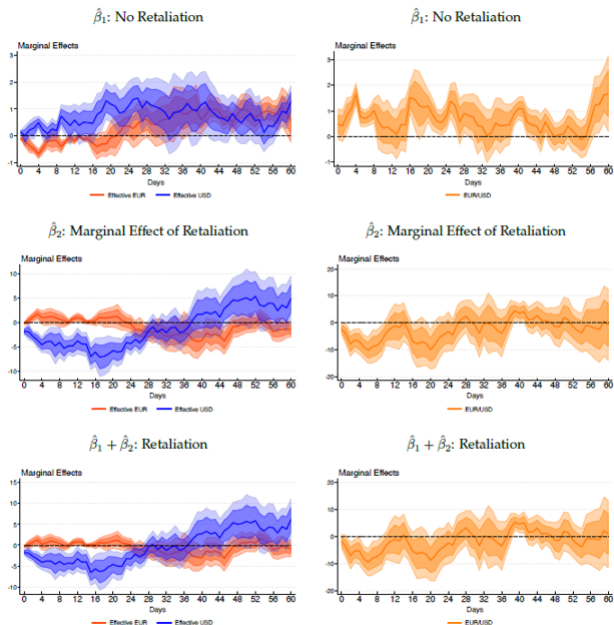
B. Impacts Conditional on Retaliation



Ostry, Daniel, Simon Lloyd, and Giancarlo Corsetti (2025). "Trading Blows: The Exchange-Rate Response to Tariffs and Retaliations", CEPR discussion paper 20452

- US-Euro: US tariff without retaliation **appreciates the effective dollar** with **limited euro impact**, also **appreciates the dollar bilaterally**.
- Reverse effect with retaliation (which more than compensates).

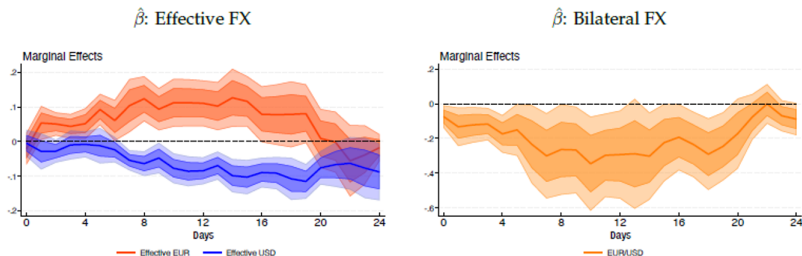
Figure 4: Exchange-Rate Impacts of 2018-2020 Global Tariff Events Conditional On Retaliation



- Similar pattern, although there has been limited retaliation.

Figure 5: Impact of 2025 Tariff Events on Asset Prices

A. Exchange Rates



Ostry, Daniel, Simon Lloyd, and Giancarlo Corsetti (2025). "Trading Blows: The Exchange-Rate Response to Tariffs and Retaliations", CEPR discussion paper 20452

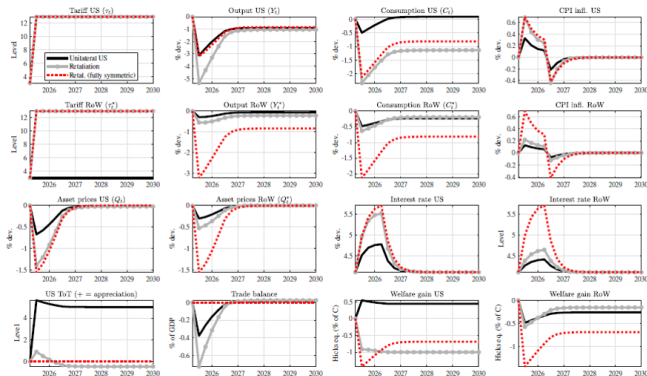
MACROECONOMIC IMPACT

- Two country DSGE model of the US and the rest of the world, with Taylor rule for monetary policy (Auray, Devereux, and Eyquem 2025).
 - US is smaller than the rest of the world, hence more exposed to trade.
- Intermediate inputs, including imported ones, used in production.
- US tariff worsens US GDP.
 - Offset by favorable change in terms-of-trade, with higher US welfare, unless there is retaliation.
- Presence of imported inputs worsens the contraction, as the tariff distorts domestic production.
- Alternative monetary policies: constant real interest rate (“look through”), full stabilization of CPI (beyond direct tariff impact).

Effect of tariff

- Unilateral US tariff reduces output (effort), with limited consumption decrease, leading to a welfare gain thanks to better terms-of-trade.
- Retaliation (if worldwide) cancels the terms-of-trade, with a high US cost as the U.S. is small relative to the world.

Figure 1: Response to an exogenous tariff shock.

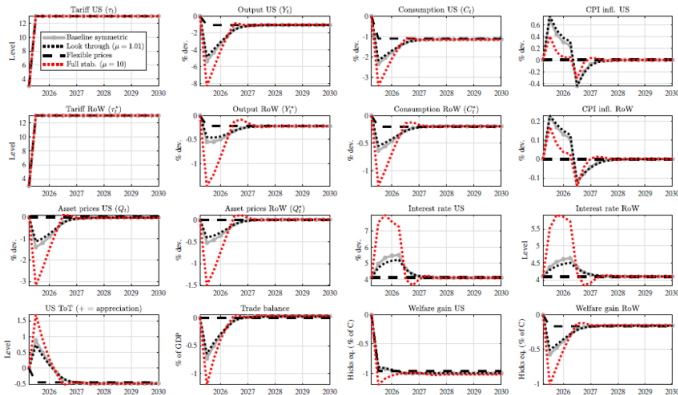


Auray, Stéphane, Michael B. Devereux, and Aurélien Eyquem (2025). "Tariffs and Retaliation: A Brief Macroeconomic Analysis", NBER working paper 33739

Monetary policy

- “Look through” policy of constant real interest rate is similar. Full CPI stabilization worsens the recession.

Figure 4: Exogenous symmetric tariff shock — Alternative monetary policies and flexible prices



Auray, Stéphane, Michael B. Devereux, and Aurélien Eyquem (2025). "Tariffs and Retaliation: A Brief Macroeconomic Analysis", NBER working paper 33739

OPTIMAL MONETARY POLICY

- How should monetary policy respond to a tariff shock (Bianchi, and Coulibaly 2025)?
- The tariff raises fiscal revenue that is rebated to households.
 - Different from a true terms-of-trade shock.
 - Households treat the rebate as exogenous, so decentralized choice differs from a planner's (fiscal externality).
- Small Open Economy (questionable for U.S.) with exogenous terms-of-trade.
- Consumption of domestic and imported goods, extension with imported intermediate.
- Standard policy with a terms-of-trade shock: target domestic (PPI) prices.
 - Leads to a decrease in employment.

- Decentralized optimality conditions differ from the social planner's optimum.
 - Allocation of consumption between imported and domestic goods.
 - Labor-consumption allocation.
- The planner treats the tariff as different from a true terms-of-trade shock, as tariff revenue is rebated.
 - If tariff revenue is not rebated, then PPI targeting is optimal.
- Decentralized choice leads to excessive drop in imports.
 - Stimulate demand and employment with an expansionary monetary policy, even though it raises inflation.
 - Employment may still drop in absolute value, but is higher than in the decentralized allocation.
- Optimal policy raises welfare, but magnitude is very small.
- Extensions:
 - Imported inputs amplify the contraction from a tariff.
 - Serie of small economies to get endogenous terms-of-trade.

Effect of tariff

- Contraction, but by less if monetary policy is optimally expansionary. Extra inflation (0.2%).

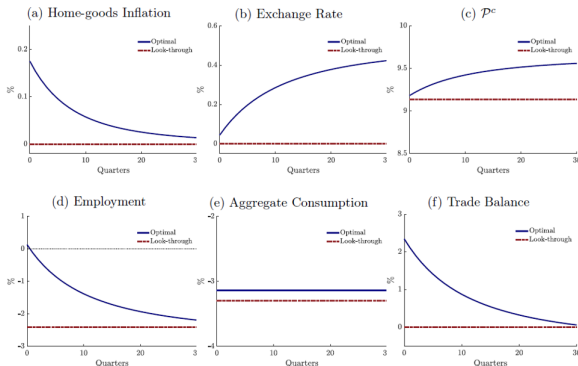


Figure 2: Baseline response to a permanent tariff

Notes: The tariff is set to $\tau_t = 15\%$ for all t . Inflation is annualized. The trade balance and the NFA position are expressed as a fraction of GDP. Consumption, employment, the exchange rate, and the tradable consumer price level, $\mathcal{P}^c$, are expressed in percentage deviation from the pre-tariff allocation.

Bianchi, Javier, and Loughou Coulibaly (2025). "The Optimal Monetary Policy Response to Tariffs", NBER working paper 33560

Imported inputs

- Similar pattern, but with higher magnitude (extra 0.4% inflation).

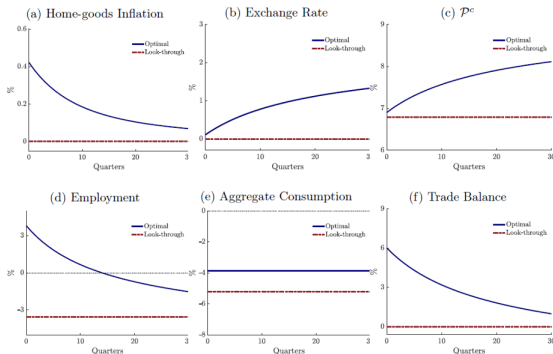


Figure 3: Response to a tariff shock in the model with imported inputs.

Notes: The tariff is imposed on imported consumption goods and intermediate inputs with $\tau_t = 15\%$ for all t . Inflation is annualized. The trade balance and the NFA position are expressed as a fraction of GDP. Consumption, employment, the exchange rate, and the tradable consumer price level, $\mathcal{P}^c$, are expressed in percentage deviation from the pre-tariff allocation.

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Welfare gains

- Optimal policy increases welfare, but only very marginally.

Table 2: Welfare Implications

	Gains Optimal Policy	Losses from Tariffs	
		Optimal Policy	Look-through
Baseline	0.009	0.99	1.00
Anticipated tariffs	0.008	0.96	0.97
Temporary tariffs	0.001	0.19	0.19
Endogenous TOT	0.007	0.68	0.69
Model w/ imported inputs			
Tariffs on c and x	0.32	1.61	1.91
Tariffs on c	0.01	1.00	1.01
Tariffs on x	0.22	0.59	0.80

Note: Welfare corresponds to permanent consumption equivalence and is expressed in percentage.

Bianchi, Javier, and Louphou Coulibaly (2025). "The Optimal Monetary Policy Response to Tariffs", NBER working paper 33560